

Binary Runs in Rueppel Hankel Determinants

Four conjectures, a polynomial refinement,
and a correction

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19 September 2026

Abstract

Let $r(x) = \sum_{j \geq 0} x^{2^j-1}$, and let $H_n(f)$ denote the Hankel determinant of order n of the coefficients of f . We determine, as a polynomial in t , every determinant of

$$f_t(x) = tx + \frac{1}{r(x^2)}.$$

If $\mathcal{R}(n)$ is the number of binary runs of n and $\varepsilon(n)$ is the lowest bit of its binary reflected Gray code, the answer is

$$H_n(f_t) = (-1)^{1+\varepsilon(n)+\lfloor \mathcal{R}(n)/2 \rfloor} (\varepsilon(n) + (\mathcal{R}(n) - \varepsilon(n))t^2), \quad n \geq 1.$$

In particular, $|H_n(f_1)| = \mathcal{R}(n)$. Together with reciprocal-minor and parity-block identities, this proves the statements numbered 8, 10, 13, and 23 in Paul Barry's 2021 note on generalized Rueppel sequences. We also classify all exceptional complex parameters, compute exact signed distributions and dyadic sums, and give a counterexample to Conjecture 22 as printed, with a valid signed replacement. The proofs are algebraic and apply to every index. An accompanying standard-library Python implementation independently checks 1,192 Rueppel-related determinants and supplies reproducible data.

Research status. This is an unrefereed, independently developed manuscript, not a claim of established publication priority. The selected statements retain their conjecture labels in the checked 2021 source. A focused search did not locate a later resolution, but does not exclude one. Related foundational identities are already present in earlier work; the source comparison and the limits of the novelty claim are made explicit in Sections 1 and 9.

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1 The problem and the main result

A sequence supported only at the numbers $2^j - 1$ seems an unlikely source of complicated determinants. Nevertheless, taking a reciprocal of its generating function produces rapidly growing signed coefficients, while its Hankel determinants retain a remarkably small, digit-controlled structure. The main problem studied here is to identify that structure and to explain why it survives the reciprocal operation.

Write

$$r_n = [n + 1 \text{ is a power of } 2], \quad r(x) = \sum_{n \geq 0} r_n x^n = 1 + x + x^3 + x^7 + x^{15} + \cdots \quad (1.1)$$

This is the Rueppel sequence, [A036987](#) [4]. The functional equation

$$r(x) = 1 + xr(x^2) \quad (1.2)$$

is immediate from its support. In Barry's note [1], Conjecture 13 asks whether the Hankel transform of $x + 1/r(x^2)$ is a signed version of the binary-run sequence [A005811](#), shifted by one place. We make the sign explicit and evaluate a whole polynomial family.

For a power series $f(x) = \sum_{j \geq 0} f_j x^j$, our convention is

$$H_n(f) = \det(f_{i+j})_{0 \leq i, j < n}, \quad H_0(f) = 1. \quad (1.3)$$

Thus n is a matrix size. Barry's Hankel-transform index is one less: his term with index j is our H_{j+1} . This convention matters in every comparison below.

For $n > 0$, let $\mathcal{R}(n)$ count the maximal runs of equal digits in the usual binary expansion of n , with no leading zeros. Set $\mathcal{R}(0) = 0$. Let

$$g(n) = n \text{ xor } \lfloor n/2 \rfloor, \quad \varepsilon(n) = g(n) \bmod 2. \quad (1.4)$$

Here $g(n)$ is the binary reflected Gray code of n . Equivalently, $\varepsilon(n) = 1$ for $n \equiv 1, 2 \pmod{4}$, and it is zero otherwise.

Theorem 1.1 (Polynomial refinement of the binary-run conjecture). *For every $n \geq 1$, the following identity holds in $\mathbb{Z}[t]$:*

$$H_n\left(tx + \frac{1}{r(x^2)}\right) = (-1)^{1+\varepsilon(n)+\lfloor \mathcal{R}(n)/2 \rfloor} (\varepsilon(n) + (\mathcal{R}(n) - \varepsilon(n))t^2). \quad (1.5)$$

The empty determinant is $H_0 = 1$, separately. Consequently, the displayed identity remains valid after any specialization of t in a commutative ring.

The proof is given in Section 5. It uses a rank-two block calculation, not induction on large reciprocal coefficients.

Corollary 1.2 (Conjecture 13). *For $f_1(x) = x + 1/r(x^2)$,*

$$|H_n(f_1)| = \mathcal{R}(n) \quad (n \geq 1). \quad (1.6)$$

In the source's indexing, the absolute value of term j is [A005811](#)($j + 1$).

Proof. At $t = 1$, the parenthesis in (1.5) is $\mathcal{R}(n) > 0$. □

1.1 Scope of the result

The targeted statements are from a single source, rather than a collection of conjectures synthesized after observing the answer. The source's conjecture numbers and our conclusions are as follows.

Number	Subject	Conclusion here
8	Binary-run parity in $H_{j+1}(1/(1 + xr(x)))$	Exact sign formula; proved.
10	Hankel transform of $1 - x/r(x^2)$	Period 4; proved.
13	Hankel transform of $x + 1/r(x^2)$	Full polynomial refinement; proved.
23	Product of unshifted and shifted Rueppel determinants	Exact binary-run parity; proved.
22	Complement and first-difference determinants	False as printed; signed replacement proved.

Allouche, Han, and Shallit [2] had already proved conjectures from Barry's earlier 2020 paper [3], including results for $1 \pm xr(x)$ and recurrences for shifted Rueppel determinants. Those results are part of the prior framework, not claimed as discoveries here. We rederive the required ingredients so that the present argument can be checked without importing their proofs. The four conjecture numbers above refer specifically to the 2021 paper, not to the same numbers in the earlier paper.

2 Binary statistics and formal-series conventions

All series are formal. No analytic convergence or choice of an inverse function is involved in the expression $1/r(x^2)$: its constant term is one, so its multiplicative reciprocal exists uniquely over $\mathbb{Z}$.

It is useful to distinguish two digit statistics. We write $\mathcal{R}(n)$ for the total number of constant-bit runs, and $\rho(n)$ for the number of runs consisting of ones. Thus, for example,

$$10 = (1010)_2, \quad \mathcal{R}(10) = 4, \quad \rho(10) = 2.$$

We use the empty expansion at zero, so both statistics vanish there. The sequences $\mathcal{R}(n)$ and $\rho(n) \bmod 2$ are respectively [A005811](#) and [A268411](#) [5, 6].

Lemma 2.1 (Digit identities). *For $m \geq 0$,*

$$\mathcal{R}(2m) = \mathcal{R}(m) + (m \bmod 2), \tag{2.1}$$

$$\mathcal{R}(2m + 1) = \mathcal{R}(m) + 1 - (m \bmod 2), \tag{2.2}$$

$$\rho(2m) = \rho(m), \tag{2.3}$$

$$\rho(2m + 1) = \rho(m) + 1 - (m \bmod 2). \tag{2.4}$$

Moreover,

$$\mathcal{R}(n) = \text{popcount}(g(n)), \quad \rho(n) = \lceil \mathcal{R}(n)/2 \rceil. \tag{2.5}$$

If

$$a(n) = \text{popcount}(\lfloor g(n)/2 \rfloor), \quad b(n) = g(n) \bmod 2, \tag{2.6}$$

then

$$b(n) = \varepsilon(n), \quad a(n) = \mathcal{R}(n) - \varepsilon(n) = \mathcal{R}(\lfloor n/2 \rfloor), \quad \binom{n}{2} \equiv a(n) \pmod{2}. \tag{2.7}$$

Proof. Appending a zero creates a new run exactly when the previous last digit is one. Appending a one creates a new run of ones, and a new total run, exactly when the previous last digit is zero; the same statement gives the stipulated value at $m = 0$. This proves the first four identities.

Write the binary digits as n_i , with all sufficiently large $n_i = 0$. The i th Gray-code digit is $n_i \text{ xor } n_{i+1}$. A one in that code records a change between adjacent binary digits, including the change from the leading one to the next higher zero. Its number of ones therefore equals the number of binary runs. Since the first run is a run of ones, runs of ones and zeros alternate, and there are $\lceil \mathcal{R}(n)/2 \rceil$ runs of ones.

Right shift commutes with the Gray-code map. Deleting the lowest Gray-code digit thus leaves the Gray code of $\lfloor n/2 \rfloor$, proving the first three assertions of (2.7). Finally, $\binom{n}{2} \bmod 2$ is the second-lowest binary digit n_1 . The xor of all Gray-code digits with indices at least one telescopes to n_1 . That xor is the parity of their number of ones, namely $a(n)$. $\square$

For shifted Hankel determinants we use

$$H_n^{[s]}(f) = \det(f_{i+j+s})_{0 \leq i, j < n}, \quad f_j = 0 \text{ when } j < 0. \quad (2.8)$$

Every determinant of size zero is one, whatever the shift. The zero-extension convention in (2.8) will be needed for $s = -1$.

3 Three determinant tools

3.1 One coupling between two blocks

If A is a nonempty square matrix, let A° mean the matrix obtained by deleting its first row and first column. In particular, for a 1×1 matrix, $\det A^\circ = 1$.

Lemma 3.1 (Rank-two coupling). *Let A and B be square matrices of positive sizes, and let e and f be their first coordinate vectors. For any scalar u ,*

$$\det \begin{pmatrix} A & u e f^T \\ u f e^T & B \end{pmatrix} = \det A \det B - u^2 \det A^\circ \det B^\circ. \quad (3.1)$$

Proof. A term in the permutation expansion that uses neither off-diagonal block belongs to $\det A \det B$. If a term uses one off-diagonal entry, it must also use the other: the only available crossing entries join the first coordinate of each block. These two entries form a transposition, contributing $-u^2$; the remaining choices are exactly the two deleted-block determinants. There are no other nonzero terms. This proof requires no invertibility assumption. $\square$

3.2 Square and rectangular anti-diagonal blocks

Lemma 3.2 (Parity-block determinants). *If C is $m \times m$ and A is any matrix of that size, then*

$$\det \begin{pmatrix} A & C \\ C^T & 0 \end{pmatrix} = (-1)^m (\det C)^2. \quad (3.2)$$

If C is $(m+1) \times m$, $e = (1, 0, \dots, 0)^T$, and C_ is obtained by deleting the first row of C , then*

$$\det \begin{pmatrix} u e e^T & C \\ C^T & 0 \end{pmatrix} = u (-1)^m (\det C_*)^2. \quad (3.3)$$

The second formula includes $m = 0$.

Proof. In (3.2), the last m rows must use the first m columns. Consequently the first m rows use the last m columns, so A contributes nothing. Exchanging the two column blocks gives the sign $(-1)^{m^2} = (-1)^m$ and the product $\det C \det C^T$.

In (3.3), there is one more upper row than lower column. A nonzero permutation term must therefore use one entry from the upper-left block. Its only possible entry is the first diagonal entry u . Removing that row and column leaves the square-block case with C_* . The corresponding cofactor has positive sign. $\square$

3.3 Complementary minors of reciprocal series

The next identity is useful beyond the Rueppel sequence. Its negative shift is the reason the apparently unrelated series $xr(x)$ enters the main calculation.

Lemma 3.3 (Reciprocal Hankel minors). *Let $A(x) = \sum_{j \geq 0} a_j x^j$ have $a_0 = 1$, and put $C(x) = 1/A(x)$. For $n \geq 0$ and $s \geq 1$,*

$$H_n^{[s]}(C) = (-1)^{n+\binom{s-1}{2}} H_{n+s-1}^{[2-s]}(A). \quad (3.4)$$

Also, for $n \geq 1$,

$$H_n(C) = (-1)^{n-1} H_{n-1}^{[2]}(A). \quad (3.5)$$

These are polynomial identities in the coefficients of A and are valid over every commutative ring.

Proof. For a series A , form the lower triangular Toeplitz matrix $T_L(A) = (a_{i-j})_{0 \leq i, j < L}$. Its diagonal entries are one, and formal series multiplication gives

$$T_L(C) = T_L(A)^{-1}, \quad \det T_L(A) = 1.$$

We first recall, with zero-based indices, the complementary-minor identity

$$\det(M^{-1})_{I,J} = (-1)^{\sum I + \sum J} \frac{\det M_{J^c, I^c}}{\det M}. \quad (3.6)$$

Here I, J have the same size. To see it, permute the selected row and column sets to the front and apply the block inverse formula. The determinant of the relevant inverse block is the determinant of the complementary block divided by the full determinant. The row and column permutations together contribute $(-1)^{\sum I + \sum J}$. A derivation assuming invertible blocks proves the identity for generic matrices; clearing denominators gives it in full generality whenever M is invertible.

For $n \geq 1$ take $L = 2n + s - 1$ and

$$I = \{n + s - 1, \dots, 2n + s - 2\}, \quad J = \{0, \dots, n - 1\}.$$

Reversing the n columns of $T_L(C)_{I,J}$ produces $H_n^{[s]}(C)$ and contributes $(-1)^{\binom{n}{2}}$. The complementary minor of $T_L(A)$ has rows $n, \dots, 2n + s - 2$ and columns $0, \dots, n + s - 2$. Reversing its columns produces $H_{n+s-1}^{[2-s]}(A)$ and contributes $(-1)^{\binom{n+s-1}{2}}$. Furthermore, $\sum I + \sum J \equiv ns \pmod{2}$. Thus the total sign exponent is

$$\binom{n}{2} + ns + \binom{n+s-1}{2} \equiv n + \binom{s-1}{2} \pmod{2},$$

which proves (3.4) for $n \geq 1$.

When $n = 0$, the matrix on the right has zeros strictly before its anti-diagonal and ones on the anti-diagonal. Its determinant is $(-1)^{\binom{s-1}{2}}$, so the same formula gives one. Finally, apply (3.4) with $s = 2$, order $n - 1$, and the roles of A, C interchanged. This gives (3.5). $\square$

4 The Rueppel determinant backbone

Define four auxiliary determinant sequences:

$$D_n = H_n(r), \quad E_n = H_n^{[1]}(r), \quad T_n = H_n^{[2]}(r), \quad B_n = H_n^{[-1]}(r) = H_n(xr(x)). \quad (4.1)$$

All four have value one at $n = 0$. The notation B_n is used only for this auxiliary determinant, not for a power-series coefficient.

4.1 Unshifted, once-shifted, and twice-shifted determinants

Theorem 4.1. For $n \geq 0$,

$$D_n = (-1)^{\binom{n}{2}}, \quad (4.2)$$

$$E_n = (-1)^{\binom{n+1}{2} + \rho(n)}, \quad (4.3)$$

$$T_{2m} = (-1)^m, \quad T_{2m+1} = 0. \quad (4.4)$$

In addition,

$$E_n = D_{\lceil n/2 \rceil} E_{\lfloor n/2 \rfloor}. \quad (4.5)$$

Proof. The coefficient relations from (1.2) are

$$r_{2j+1} = r_j, \quad r_{2j} = [j = 0]. \quad (4.6)$$

Order both rows and columns by even indices first, then odd indices. Because the same permutation is applied to rows and columns, its two signs cancel.

For D_{2m} , the upper-left block has only its first diagonal entry nonzero, the lower-right block is zero, and the off-diagonal block is $(r_{i+j})_{0 \leq i, j < m}$. Hence

$$D_{2m} = (-1)^m D_m^2.$$

For D_{2m+1} , the off-diagonal block has one extra row. Deleting its first row leaves $(r_{i+j+1})_{0 \leq i, j < m}$, so

$$D_{2m+1} = (-1)^m E_m^2.$$

Both conclusions use Lemma 3.2.

For E_n , the off-diagonal blocks are zero. The even-even block is an unshifted Rueppel Hankel matrix of order $\lceil n/2 \rceil$, and the odd-odd block is a once-shifted one of order $\lfloor n/2 \rfloor$. This proves (4.5). Starting from $D_0 = D_1 = E_0 = E_1 = 1$, these formulas simultaneously show that all D_n, E_n are ± 1 . They then give $D_{2m} = D_{2m+1} = (-1)^m$, which is exactly (4.2). There is no circular use of nonvanishing here: at each order at least two, the formulas involve smaller orders.

To prove (4.3), it is enough to check that its right side has the same recurrence and initial value as E_n . At $n = 2m$, its exponent is congruent to $m + \rho(m)$, and the exponent of $D_m E_m$ is

$$\binom{m}{2} + \binom{m+1}{2} + \rho(m) = m^2 + \rho(m) \equiv m + \rho(m) \pmod{2}.$$

At $n = 2m+1$, the exponent of $D_{m+1} E_m$ is congruent to $\rho(m)$. Using (2.4), the proposed expression has exponent

$$(m+1)(2m+1) + \rho(m) + 1 - (m \bmod 2) \equiv \rho(m) \pmod{2}.$$

The value at zero is one, so induction proves the formula.

Finally, the even-even and odd-odd blocks for T_n are both zero. At even order $2m$, the off-diagonal block is the once-shifted matrix, with determinant E_m . Thus $T_{2m} = (-1)^m E_m^2 = (-1)^m$. At odd order, the two block sizes differ, making the matrix singular. This proves (4.4). $\square$

A frequently used consequence is

$$T_{m-1} = D_m (m \bmod 2) \quad (m \geq 1). \quad (4.7)$$

It follows directly from (4.2) and (4.4), including $m = 1$.

4.2 A negative shift detects the number of runs

Theorem 4.2. For every $n \geq 1$,

$$B_n = -E_{n-1}\mathcal{R}(n-1). \quad (4.8)$$

Consequently, for arbitrary parameters u, v ,

$$H_n(u + vxr(x)) = v^{n-1}E_{n-1}(u - v\mathcal{R}(n-1)), \quad n \geq 1. \quad (4.9)$$

Proof. Put $s_0 = 0$ and $s_j = r_{j-1}$ for $j \geq 1$, so $\sum s_j x^j = xr(x)$. Then

$$s_{2j} = s_j, \quad s_{2j+1} = [j = 0].$$

Set $e = \lceil n/2 \rceil$, $o = \lfloor n/2 \rfloor$. For $n \geq 2$, the parity-ordered matrix for B_n has diagonal blocks given by B_e and D_o , and off-diagonal blocks coupling only their first coordinates. The first deleted-block determinant is E_{e-1} and the second is T_{o-1} . Lemma 3.1 therefore gives

$$B_n = B_e D_o - E_{e-1} T_{o-1} = D_o (B_e - (o \bmod 2) E_{e-1}). \quad (4.10)$$

The base case is $B_1 = 0 = -E_0\mathcal{R}(0)$. Assume the formula holds at smaller orders. Since $e < n$ for $n \geq 2$, (4.10) becomes

$$B_n = -D_o E_{e-1} (\mathcal{R}(e-1) + (o \bmod 2)).$$

Equation (4.5) applied at $n-1$ says $E_{n-1} = D_o E_{e-1}$. The two digit recurrences (2.1)–(2.2), separated into even and odd n , give

$$\mathcal{R}(n-1) = \mathcal{R}(e-1) + (o \bmod 2).$$

This completes the induction for (4.8).

For (4.9), changing the constant term of $vxr(x)$ from zero to u changes only the first diagonal entry of its Hankel matrix. Expansion in that entry gives

$$H_n(u + vxr(x)) = v^n B_n + u v^{n-1} E_{n-1}.$$

Insert (4.8). The result is a polynomial identity, so $v = 0$ causes no division problem. At $n = 1$ the factor v^0 is one. $\square$

The specializations $u = 1, v = \pm 1$ recover the binary-run magnitudes for $1 \pm xr(x)$ that are already part of the earlier Rueppel literature. The point of the derivation is that the negative shift needed later has now been evaluated, including its sign and its value at every order.

5 Proof of the polynomial refinement

Let

$$c(x) = \frac{1}{r(x)} = 1 - x + x^2 - 2x^3 + 3x^4 - 4x^5 + 6x^6 - \dots. \quad (5.1)$$

Then $f_t(x) = c(x^2) + tx$. Only the coefficient of x among the odd-degree coefficients is nonzero. This is what makes the determinant quadratic in t , even when its matrix order is very large.

Proof of Theorem 1.1. The case $n = 1$ reads $1 = 1$, so suppose $n \geq 2$ and set $e = \lceil n/2 \rceil$, $o = \lfloor n/2 \rfloor$. Reorder by parity. The even-even block is $(c_{i+j})_{0 \leq i, j < e}$; the odd-odd block is $(c_{i+j+1})_{0 \leq i, j < o}$. The only off-diagonal coupling is t between their first coordinates. Thus

$$H_n(f_t) = H_e(c)H_o^{[1]}(c) - t^2 H_{e-1}^{[2]}(c)H_{o-1}^{[3]}(c). \quad (5.2)$$

Apply Lemma 3.3 four times. The required values are

$$H_e(c) = (-1)^{e-1}T_{e-1}, \quad H_o^{[1]}(c) = (-1)^o E_o, \quad (5.3)$$

$$H_{e-1}^{[2]}(c) = (-1)^{e-1}D_e, \quad H_{o-1}^{[3]}(c) = (-1)^o B_{o+1}. \quad (5.4)$$

The last formula also covers $o = 1$ by the size-zero part of the reciprocal-minor lemma. Substitution in (5.2) yields

$$\begin{aligned} H_n(f_t) &= (-1)^{n-1} (T_{e-1}E_o - t^2 D_e B_{o+1}) \\ &= (-1)^{n-1} D_e E_o ((e \bmod 2) + t^2 \mathcal{R}(o)) \\ &= (-1)^{n-1} E_n (\varepsilon(n) + t^2 (\mathcal{R}(n) - \varepsilon(n))). \end{aligned} \quad (5.5)$$

Here we used (4.7), (4.8), (4.5), and (2.7), respectively. This already proves the full polynomial evaluation in a useful alternative form.

It remains to express its sign using the same digit statistics as its magnitude. Formula (4.3) implies

$$(-1)^{n-1} E_n = (-1)^{\binom{n}{2} + \rho(n) + 1}.$$

Write $a = a(n)$, $b = b(n)$, so $\mathcal{R}(n) = a + b$ and $b = \varepsilon(n)$. By Lemma 2.1, the exponent is congruent to

$$1 + a + \lceil (a + b)/2 \rceil \equiv 1 + b + \lfloor (a + b)/2 \rfloor \pmod{2}.$$

This is the sign in (1.5). Every step has taken place in the polynomial ring $\mathbb{Z}[t]$, so no exceptional value of t was excluded. $\square$

Example 5.1 (Reading a determinant from its index). For $n = 10$, $g(n) = (1111)_2$, hence $\mathcal{R}(n) = 4$ and $\varepsilon(n) = 1$. The theorem gives

$$H_{10}(f_t) = 1 + 3t^2.$$

For $n = 11$, $g(n) = (1110)_2$, so $\mathcal{R}(n) = 3$ and $\varepsilon(n) = 0$, giving $H_{11}(f_t) = 3t^2$. The index $n = 10^{100}$ has $\mathcal{R}(n) = 122$ and $\varepsilon(n) = 0$, and the result is simply

$$H_{10^{100}}(f_t) = 122t^2.$$

The last digit count was computed by the supplied code; no enormous moment matrix was formed.

n	Binary n	Gray code $g(n)$	$\mathcal{R}(n)$	$\varepsilon(n)$	$H_n(f_t)$
1	1	1	1	1	1
2	10	11	2	1	$-1 - t^2$
3	11	10	1	0	$-t^2$
4	100	110	2	0	$2t^2$
5	101	111	3	1	$-1 - 2t^2$
6	110	101	2	1	$-1 - t^2$
7	111	100	1	0	$-t^2$
8	1000	1100	2	0	$2t^2$
9	1001	1101	3	1	$-1 - 2t^2$
10	1010	1111	4	1	$1 + 3t^2$
11	1011	1110	3	0	$3t^2$
12	1100	1010	2	0	$2t^2$
13	1101	1011	3	1	$-1 - 2t^2$
14	1110	1001	2	1	$-1 - t^2$
15	1111	1000	1	0	$-t^2$
16	10000	11000	2	0	$2t^2$

Table 1: The first sixteen positive-order determinant polynomials. The sign and both coefficients are read from the index, not from a recurrence in the moments.

6 Three further conjectures

6.1 Reciprocal signs: Conjecture 8

Theorem 6.1. For $n \geq 1$ and a parameter v ,

$$H_n\left(\frac{1}{1 + vxr(x)}\right) = (-v)^{n-1}E_{n-1}. \quad (6.1)$$

In particular, putting $q_j = H_{j+1}(1/(1 + xr(x)))$, we have

$$q_j = (-1)^{\binom{j}{2} + \rho(j)} \quad (j \geq 0). \quad (6.2)$$

Therefore

$$\left| \frac{(-1)^{\binom{j}{2}} - q_j}{2} \right| = \rho(j) \pmod{2}. \quad (6.3)$$

Proof. The coefficients of $1 + vxr(x)$ beginning with degree two are $vr_1, vr_2, \dots$. Formula (3.5) therefore gives

$$H_n\left(\frac{1}{1 + vxr(x)}\right) = (-1)^{n-1}v^{n-1}E_{n-1},$$

proving (6.1). At $n = j + 1$, (4.3) shows that the sign exponent is

$$j + \binom{j+1}{2} + \rho(j) \equiv \binom{j}{2} + \rho(j) \pmod{2}.$$

The two signs in the numerator of (6.3) are equal when $\rho(j)$ is even and opposite when it is odd. $\square$

The coefficients of $1/(1 + xr(x))$ form A339422 [7]. The theorem concerns their determinants, not the coefficient sequence itself. This distinction prevents a common indexing and interpretation error in numerical searches.

6.2 A product of signs: Conjecture 23

Corollary 6.2. *For all $j \geq 0$,*

$$\frac{1 + (-1)^j D_{j+1} E_{j+1}}{2} = \rho(j+1) \bmod 2. \quad (6.4)$$

Proof. With $m = j + 1$, (4.2) and (4.3) give

$$D_m E_m = (-1)^{\binom{m}{2} + \binom{m+1}{2} + \rho(m)} = (-1)^{m + \rho(m)}.$$

Hence $(-1)^j D_{j+1} E_{j+1} = -(-1)^{\rho(j+1)}$. The expression in (6.4) is one precisely when $\rho(j+1)$ is odd. $\square$

Thus Conjectures 8 and 23 are two manifestations of the same once-shifted Rueppel sign, rather than unrelated coincidences with an OEIS entry.

6.3 A periodic transform: Conjecture 10

Theorem 6.3 (A two-parameter periodic-block identity). *Let $P_{u,v}(x) = u + vx/r(x^2)$. For $m \geq 0$,*

$$H_{2m+1}(P_{u,v}) = (-1)^m u v^{2m}, \quad (6.5)$$

and, for $m \geq 1$,

$$H_{2m}(P_{u,v}) = \begin{cases} (-1)^m v^{2m}, & m \text{ odd}, \\ 0, & m \text{ even}. \end{cases} \quad (6.6)$$

The empty determinant is one.

Proof. Again let $c(x) = 1/r(x)$. The only even coefficient of $P_{u,v}$ that may be nonzero is its constant coefficient u ; its coefficient of x^{2j+1} is vc_j . At order $2m$, parity ordering gives a square off-diagonal block $v(c_{i+j})$. Lemma 3.2 gives

$$H_{2m}(P_{u,v}) = (-1)^m v^{2m} H_m(c)^2.$$

By (3.5), $H_m(c) = (-1)^{m-1} T_{m-1}$, whose square is one for odd m and zero for even m . This proves (6.6).

At order $2m + 1$, deleting the first row of the rectangular off-diagonal block leaves $v(c_{i+j+1})$. The rectangular-block identity gives

$$H_{2m+1}(P_{u,v}) = (-1)^m u v^{2m} H_m^{[1]}(c)^2.$$

By (3.4), $H_m^{[1]}(c) = (-1)^m E_m$, whose square is one. This also handles $m = 0$. $\square$

Corollary 6.4 (Conjecture 10). *The Hankel transform, with source index starting at zero, of $1 - x/r(x^2)$ is*

$$1, -1, -1, 0, 1, -1, -1, 0, 1, -1, -1, 0, \dots$$

Its generating function is

$$\sum_{j \geq 0} H_{j+1} \left(1 - \frac{x}{r(x^2)} \right) z^j = \frac{1 - z - z^2}{1 - z^4}. \quad (6.7)$$

Proof. Specialize $u = 1, v = -1$ in Theorem 6.3 and separate the four residue classes of the matrix order. The displayed generating function is the sum of the resulting period-four blocks. $\square$

7 Consequences of the polynomial formula

7.1 The exact singular-parameter set

One reason to retain t instead of specializing immediately is that (1.5) answers a global nonvanishing question at no additional determinant cost.

Theorem 7.1 (All exceptional complex parameters). *For a fixed $t \in \mathbb{C}$, every leading Hankel determinant of f_t is nonzero if and only if*

$$t \notin \{0\} \cup \left\{ \frac{i}{\sqrt{k}}, -\frac{i}{\sqrt{k}} : k = 1, 2, 3, \dots \right\}. \quad (7.1)$$

Each excluded parameter makes infinitely many positive-order determinants vanish. In particular, every real $t \neq 0$ gives nonzero determinants of every order.

Proof. Use $a = a(n)$ and $b = b(n)$ from (2.6). Up to a sign, the polynomial is $b + at^2$. For $n \geq 1$, $(a, b) \neq (0, 0)$. If $b = 0$, then $a \geq 1$, and the only root is $t = 0$. If $b = 1$ and $a = 0$, the polynomial is one. If $b = 1$ and $a = k \geq 1$, its roots are precisely $\pm i/\sqrt{k}$.

It remains to show that none of these possibilities is merely formal. Every nonnegative integer occurs as a Gray code: given its digits, recover the binary digits by cumulative xor from the highest position. For each $k \geq 1$, infinitely many Gray codes have lowest bit one and exactly k ones above that bit. They yield infinitely many indices with the factor $1 + kt^2$. Similarly, infinitely many nonzero Gray codes have lowest bit zero, yielding roots at $t = 0$. The converse follows directly from the exhaustive cases for $b + at^2$. $\square$

Nonvanishing here is not positivity. For example, $H_2(f_t) = -1 - t^2 < 0$ for real t , so these moments do not define a positive-definite real Hankel form. The result is an exact nondegeneracy statement.

7.2 Polynomial and signed-value distributions

Theorem 7.2 (Dyadic polynomial distribution). *Fix $N \geq 1$. Among indices $1 \leq n < 2^N$, the pair $(a(n), b(n)) = (k, b)$, where $0 \leq k \leq N - 1$ and $b \in \{0, 1\}$, occurs*

$$\binom{N-1}{k} - [k = 0, b = 0] \quad (7.2)$$

times. All indices with that pair have the same determinant polynomial, namely

$$H_n(f_t) = \begin{cases} (-1)^{1+\lfloor k/2 \rfloor} kt^2, & b = 0, \\ (-1)^{\lfloor (k+1)/2 \rfloor} (1 + kt^2), & b = 1. \end{cases} \quad (7.3)$$

Proof. The Gray-code map permutes the N -bit strings. With the lowest bit fixed, there are $\binom{N-1}{k}$ ways to choose the k higher ones. The all-zero string corresponds to the excluded index $n = 0$. The two signs in (7.3) are the main theorem's sign, written separately for $b = 0$ and $b = 1$. $\square$

A compact form of this distribution is the bivariate identity

$$\sum_{1 \leq n < 2^N} u^{a(n)} v^{b(n)} = (1 + v)(1 + u)^{N-1} - 1. \quad (7.4)$$

It describes the constant and quadratic coefficients before the explicit sign is attached; it is not a generating function of the original power-series coefficients.

Corollary 7.3 (Exact signed magnitudes at $t = 1$). For $1 \leq k \leq N$, put $s_k = (-1)^{\lfloor k/2 \rfloor}$. Among $1 \leq n < 2^N$,

$$\#\{n : H_n(f_1) = s_k k\} = \binom{N-1}{k-1}, \quad (7.5)$$

$$\#\{n : H_n(f_1) = -s_k k\} = \binom{N-1}{k}. \quad (7.6)$$

In particular, $|H_n(f_1)| = k$ occurs $\binom{N}{k}$ times.

Proof. For magnitude k , the case $b = 1$ has $a = k - 1$ and sign s_k ; the case $b = 0$ has $a = k$ and the opposite sign. Add the two counts and use Pascal's identity. Binomial coefficients outside their usual range are understood to be zero. $\square$

The largest magnitude on this interval is N , attained once, at the index with all N Gray-code digits equal to one. Its binary digits alternate, and its value is

$$n = \lfloor 2^{N+1}/3 \rfloor, \quad H_n(f_1) = (-1)^{\lfloor N/2 \rfloor} N. \quad (7.7)$$

Thus the determinants grow at most logarithmically in their matrix order, despite the much larger coefficients from which they are built.

7.3 Dyadic sums with the parameter retained

Theorem 7.4. For $N \geq 2$ and arbitrary t ,

$$\sum_{1 \leq n < 2^N} H_n(f_t) = \operatorname{Re}(1+i)^N - 2(N-1)t^2 \operatorname{Re}(1+i)^{N-2}. \quad (7.8)$$

At $N = 1$ the sum is one. For real t and all $N \geq 1$,

$$\sum_{1 \leq n < 2^N} |H_n(f_t)| = 2^{N-1} (1 + (N-1)t^2). \quad (7.9)$$

The real parts in (7.8) are integers; the identity is an identity of polynomials in t .

Proof. Set $M = N - 1$. Let $\sigma_0(k) = (-1)^{1+\lfloor k/2 \rfloor}$ and $\sigma_1(k) = (-1)^{\lfloor (k+1)/2 \rfloor}$. Their four-periodic patterns show that

$$\sigma_1(k) = \operatorname{Re}((1+i)i^k), \quad \sigma_0(k) + \sigma_1(k) = -2 \operatorname{Im}(i^k).$$

The missing pair $(0, 0)$ contributes zero, so Theorem 7.2 gives a constant coefficient

$$\sum_{k=0}^M \binom{M}{k} \sigma_1(k) = \operatorname{Re}((1+i)(1+i)^M) = \operatorname{Re}(1+i)^N.$$

The coefficient of t^2 is

$$\begin{aligned} -2 \operatorname{Im} \sum_{k=0}^M k \binom{M}{k} i^k &= -2 \operatorname{Im}(M i (1+i)^{M-1}) \\ &= -2M \operatorname{Re}(1+i)^{M-1}. \end{aligned}$$

This proves (7.8) when $M \geq 1$; the case $N = 1$ is direct.

For real t , the unsigned polynomial is $b + at^2 \geq 0$. Summing over all pairs gives

$$\sum_{k=0}^M \binom{M}{k} (kt^2 + (1+kt^2)) = 2^M + M2^M t^2,$$

using $\sum k \binom{M}{k} = M2^{M-1}$ for $M \geq 1$. The case $M = 0$ is again immediate. This is (7.9). $\square$

8 A printed counterexample and a signed replacement

The four affirmative results above are the principal conclusions. There is also a small but important correction in the same source. We treat it separately because finding an omitted sign or absolute value is not comparable in significance to a structural theorem.

8.1 What fails in Conjecture 22

Let

$$C_m = \det(1 - r_{i+j})_{0 \leq i, j < m}, \quad J_m = \det(r_{i+j+1} - r_{i+j})_{0 \leq i, j < m}. \quad (8.1)$$

The formula printed as Conjecture 22 would imply, in matrix-size notation,

$$J_m^2 = |C_{m+1}| - |C_m| \quad (m \geq 1). \quad (8.2)$$

At $m = 3$, however,

$$C_3 = \det \begin{pmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{pmatrix} = -1, \quad C_4 = \det \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & 1 \end{pmatrix} = 0.$$

The last row of the second matrix is the sum of its first two rows. On the other hand,

$$J_3 = \det \begin{pmatrix} 0 & -1 & 1 \\ -1 & 1 & -1 \\ 1 & -1 & 0 \end{pmatrix} = 1.$$

Thus (8.2) asserts $1 = -1$. In the source's indexing, this is the counterexample $j = 2$. Direct checks at $m = 1, 2$ satisfy the formula, so this is its first failure. The assertion is therefore false as printed; an additional absolute value or a different sign convention would constitute a different assertion.

8.2 A general moment identity

There is a natural exact replacement that avoids guessing which signs to suppress. In fact it applies to arbitrary moment sequences.

Theorem 8.1 (Complement–difference square identity). *Let $(a_j)_{j \geq 0}$ be any sequence, and define*

$$d_m = \det(a_{i+j}), \quad c_m = \det(1 - a_{i+j}), \quad j_m = \det(a_{i+j+1} - a_{i+j}),$$

where all matrices have order m . Then for $m \geq 1$,

$$j_m^2 = (-1)^m (d_m c_{m+1} + d_{m+1} c_m). \quad (8.3)$$

This polynomial identity requires no nonvanishing assumptions on the determinants. In the Rueppel case it becomes

$$J_m^2 = D_m C_m + D_{m+1} C_{m+1}. \quad (8.4)$$

Proof. We first work with generic moments over a rational-function field, where the needed inverses exist. Put

$$A_m = (a_{i+j})_{0 \leq i, j < m}, \quad v = (a_m, \dots, a_{2m-1})^T, \quad K_m = \mathbf{1}^T A_m^{-1} \mathbf{1}.$$

The matrix determinant lemma gives

$$c_m = (-1)^m d_m (1 - K_m). \quad (8.5)$$

The next Hankel matrix is

$$A_{m+1} = \begin{pmatrix} A_m & v \\ v^T & a_{2m} \end{pmatrix},$$

whose Schur complement is $s = a_{2m} - v^T A_m^{-1} v = d_{m+1}/d_m$. Its block inverse gives

$$K_{m+1} - K_m = \frac{(1 - v^T A_m^{-1} \mathbf{1})^2}{s}. \quad (8.6)$$

To identify the numerator, consider the bordered matrix

$$M = \begin{pmatrix} a_0 & a_1 & \cdots & a_m \\ a_1 & a_2 & \cdots & a_{m+1} \\ \vdots & \vdots & & \vdots \\ a_{m-1} & a_m & \cdots & a_{2m-1} \\ 1 & 1 & \cdots & 1 \end{pmatrix}.$$

Its determinant by a Schur complement is $d_m(1 - v^T A_m^{-1} \mathbf{1})$. Alternatively, for columns $m, m-1, \dots, 1$, subtract the preceding column. The last row becomes $(1, 0, \dots, 0)$; expanding there leaves the first-difference Hankel determinant with sign $(-1)^m$. Therefore

$$d_m(1 - v^T A_m^{-1} \mathbf{1}) = (-1)^m j_m.$$

Substitution into (8.6) yields

$$K_{m+1} - K_m = \frac{j_m^2}{d_m d_{m+1}}.$$

Now use (8.5) at the two consecutive orders and rearrange:

$$j_m^2 = (-1)^m (d_m c_{m+1} + d_{m+1} c_m).$$

Both sides are polynomials with integer coefficients in finitely many moments. Equality for generic moments proves the polynomial identity, so specialization is valid even when some determinants vanish and over any commutative ring.

For the Rueppel sequence, $D_{m+1} = (-1)^m D_m$. Inserting this relation into (8.3) gives (8.4). $\square$

At the counterexample above, the corrected formula reads $J_3^2 = D_3 C_3 + D_4 C_4 = (-1)(-1) + (1)(0) = 1$, as required. We do not claim that simply adding an outer absolute value to the printed formula has been proved here; the precise replacement established is the signed polynomial identity (8.4).

9 Reproducibility, status, and remaining questions

9.1 What was checked computationally

The mathematical proofs above apply at every order. Computation serves as an independent audit of the indexing, signs, polynomial specializations, and implementation; it is not a substitute for those proofs.

The archive contains a standard-library Python module, a separate test suite, an exact verification script, and the outputs of an actual run. The determinant engine is fraction-free Bareiss elimination with row pivoting. Every division is checked for exactness. Thus neither a small floating-point residual nor an approximate zero is accepted as evidence. The engine does not call any of the binary formulas when evaluating a matrix determinant.

Audit	Executed scope
Rueppel-related determinants	1,192 exact determinant evaluations.
Consecutive orders	All orders 0 through 64 for each tested family.
Selected larger orders	65, 127, 128 for the seven indicated base families.
Parameter values for f_t	$t = -2, 0, 1, 2$.
General square identity	500 checks on randomly generated integer moments.
Digital recurrence for E_n	Every $0 \leq n < 2^{16}$, plus 1,000 random 256-bit indices.
Dyadic distributions and sums	Distribution at $N = 16$; sums for $1 \leq N \leq 16, t = 0, 1, 2$.
Independent determinant audit	Leibniz permutation sums on 140 random matrices, sizes 0 through 6.
Reciprocal-minor identity audit	Random integer series and shifts 1 through 5, including negative complementary shifts.

The seven families receiving the larger-order checks are $D, E, T, B, f_1, 1 - x/r(x^2)$, and $1/(1 + xr(x))$. Complement determinants were also computed at order 65 to check the square identity through order 64. The complete verification run used Python 3.13.5 and took about 7.29 seconds in the execution environment; this is a record of that run, not a portable timing guarantee. All checks passed.

Two implementation details are particularly important. First, the size-zero determinant is handled explicitly and is never confused with the first term of a Hankel transform. Second, singular matrices and zero leading pivots are tested intentionally. The periodic family contains infinitely many zero determinants, so a routine that silently assumes nonzero pivots would be unsuitable.

From the archive directory, the two reproduction commands are

```
python -m unittest discover -s tests -v
python code/verify.py
```

The default verification output overwrites files under `data/`. Larger finite audits can be requested, for example,

```
python code/verify.py --max-size 96 --large-size 192 --digital-bits 18
```

This last command is an optional example, not a run claimed in the reported results. No external Python packages, internet access, Wolfram kernel, or proof assistant are required by the supplied code.

9.2 Source and novelty audit

The primary problem source is [arXiv:2107.00442v2](#), dated 4 July 2021. The statements and formulas were checked against the PDF, specifically printed pages 6–8 and 14, as well as the HTML representation. The arXiv abstract page’s current version history was checked on 19 September 2026. The earlier paper by Allouche, Han, and Shallit was consulted through [arXiv:2006.08909v2](#); its journal version is listed in the bibliography. The relevant OEIS entries were also checked on that date.

The present proofs establish the stated identities mathematically, subject to ordinary independent scrutiny of a new manuscript. They do not establish bibliographic priority. In particular,

the evaluation of D_n , recurrences for E_n , and binary-run phenomena for $1 \pm xr(x)$ are prior material. The parameter formula, the explicit deductions of the four targeted 2021 assertions, and their collected consequences are the outputs of this investigation. A focused search did not identify a later paper resolving those exact assertions, but unindexed proofs, unpublished communications, or equivalent formulations may exist. No claim is made that an OEIS record or the author's paper has been updated to reflect this manuscript.

The correction to Conjecture 22 is deliberately described as a counterexample to the *printed statement*. Its small size makes a typographical or sign-convention explanation plausible; nothing here purports to infer the author's intended corrected formula.

9.3 Boundaries of the result

The proofs do not resolve every conjecture in the 2021 paper. In particular, no conclusion is asserted here about its other parameterized families or its period-eight assertion modulo two.

The method suggests two concrete extensions. One is to evaluate higher shifts $H_n^{[s]}(r)$ uniformly in both n and s , which would make the reciprocal-minor lemma more broadly effective. Another is to replace the perturbation tx by a polynomial with several odd-degree terms. Parity ordering would then produce a finite-rank coupling, but its minors would require more than the four determinant sequences used here. These are directions for further work, not results asserted by this manuscript.

A Direct evaluation and complexity

The main formula is not merely a recurrence. It is an explicit function of the binary digits of the matrix order. The following code returns the constant and quadratic coefficients of the determinant polynomial. It is also included, with input checks and the other family formulas, in `code/rueppel.py`.

```
def parameter_coefficients(n: int) -> tuple[int, int]:
    if not isinstance(n, int) or isinstance(n, bool) or n < 0:
        raise ValueError("The index must be a nonnegative integer.")
    if n == 0:
        return 1, 0
    gray = n ^ (n >> 1)
    a = (gray >> 1).bit_count()
    b = gray & 1
    exponent = 1 + b + (a + b) // 2
    sigma = -1 if exponent % 2 else 1
    return sigma * b, sigma * a
```

If the input index has L binary digits, xor, shifting, and population count need $O(L)$ bit work in a straightforward implementation. The output coefficients have $O(\log L)$ bits. Thus the determinant polynomial can be evaluated using work linear in the input bit length, without generating moments or building an $n \times n$ matrix. The displayed Python implementation stores intermediate integers of $O(L)$ bits; it is not a claim of constant-space computation for arbitrary-precision inputs.

For comparison, the independent matrix audit uses a cubic number of integer arithmetic operations per determinant, with bit cost depending on the intermediate entries. Its purpose is verification from the definition. Replacing it by the binary formula inside the audit would make the main comparison circular.

B Archive contents

File	Purpose
rueppel_hankel.tex, tables.tex	Complete article source and generated table.
rueppel_hankel.pdf	Compiled article.
code/rueppel.py	Exact formulas, formal reciprocals, and determinant engine.
code/verify.py	Reproduction script for the recorded experiments.
code/make_table.py	Regenerates the first-values table.
tests/test_rueppel.py	Independent small-size and identity tests.
data/exact_determinants.csv	Direct matrix evaluations through order 64.
data/polynomial_coefficients.csv	Binary data and determinant polynomials through order 1024.
data/dyadic_distribution.csv	The coefficient distribution at $N = 16$.
data/dyadic_sums.json	Exact sums for the tested dyadic ranges.
data/verification_summary.json	Machine-readable audit summary and counterexample.
data/verification_log.txt	Output of the executed verification run.
README.md, Makefile	Reading and build instructions.
SOURCE_AUDIT.md	Source locations and priority caveats.

References

- [1] Paul Barry, *Conjectures and results on some generalized Rueppel sequences*, [arXiv:2107.00442v2](#), 4 July 2021. [doi:10.48550/arXiv.2107.00442](#).
- [2] Jean-Paul Allouche, Guo-Niu Han, and Jeffrey Shallit, *On some conjectures of P. Barry*, *Journal of Number Theory* **228** (2021), 108–132. [doi:10.1016/j.jnt.2021.03.015](#). Preprint: [arXiv:2006.08909v2](#), 24 June 2020.
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